

## Working with Selenium 2

practiceautomation.com

**Step 1:** create a maven project on terminal by typing the command **mvn**

**archetype:generate -DgroupId=com.example -**

**DartifactId=MyMavenPracticeAutomation -DarchetypeArtifactId=maven-**

**archetype-quickstart -DinteractiveMode=false**

```
samiksha@samiksha:~/Devops00$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenPracticeAutomation -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.456 s
[INFO] Finished at: 2026-03-17T20:40:07+05:30
[INFO] -----
```

**Step 2 :** edit the pom.xml file and add the dependencies

The pom.xml has the code

```
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenPracticeAutomationApp</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>MyMavenPracticeAutomationApp</name>
  <url>http://maven.apache.org</url>

  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>

    <!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
    <dependency>
      <groupId>org.seleniumhq.selenium</groupId>
      <artifactId>selenium-java</artifactId>
      <version>4.29.0</version>
    </dependency>
  </dependencies>
```

```

    <!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
    <groupId>org.apache.maven.plugins</groupId>

    <artifactId>maven-jar-plugin</artifactId>
    <version>3.0.1</version>
    <configuration>
        <archive>
            <manifestEntries>
                <Main-Class>com.example.App</Main-Class>
            </manifestEntries>
        </archive>
    </configuration>
</plugin>

</plugins>
</build>
</project>

```

---

### Step 3: Edit the App.java

```

package com.example;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
public class App
{
    public static void main( String[] args )
    {
        WebDriver driver = new ChromeDriver();
        driver.get("https://practicetestautomation.com/practice-test-login/");
        driver.manage().window().maximize();
        driver.findElement(By.id("username")).sendKeys("student");
        driver.findElement(By.id("password")).sendKeys("Password123");
        driver.findElement(By.id("submit")).click();
    }
}

```

### Step 4: build and run the maven application

To execute type the command **mvn exec:java -Dexec.mainClass="com.example.App"** it opens practiceautomation.com and we can login using the credentials



# Logged In Successfully

Congratulations student. You successfully logged in!

[Log out](#)

**Step 5:** now open the terminal and type the command **cd MyMavenPracticeAutomationApp** to enter the folder and create a **Jenkinsfile** .Paste the code in the Jenkinsfile

```
pipeline {
  agent any // Use any available agent
  tools {
    maven 'Maven' // Ensure this matches the name configured in Jenkins
  }
  stages {
    stage('Checkout') {
      steps {
        git branch: 'master', url:
'https://github.com/Samiksha201104/MyMavenPracticeAutomationApp.git'
      }
    }
    stage('Build') {
      steps {
        sh 'mvn clean package' // Run Maven build
      }
    }
    stage('Test') {
```

```

    steps {
        sh 'mvn test' // Run unit tests
    }
}

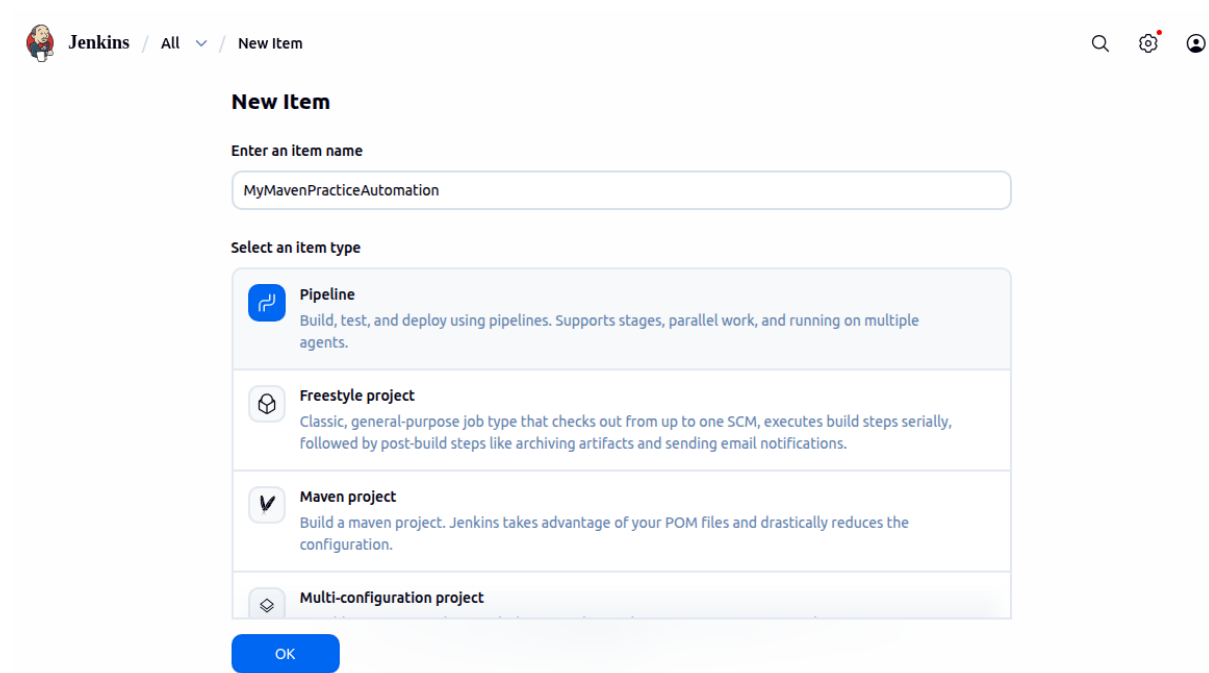
stage('Run Application') {
    steps {
        // Start the JAR application
        sh 'java -jar target/MyMavenApp-1.0-SNAPSHOT.jar'
    }
}
}

post {
    success {
        echo 'Build and deployment successful!'
    }
    failure {
        echo 'Build failed!'
    }
}
}
}

```

Then commit and push it to GitHub

**Step 6:** in the Jenkins Dashboard click on new item type the name of the item and select Pipeline



Jenkins / All / New Item

**New Item**

Enter an item name

MyMavenPracticeAutomation

Select an item type

- Pipeline**  
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project**  
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Maven project**  
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.
- Multi-configuration project**

OK

In configure scroll down to Pipeline and choose pipeline script from SCM, give the repository url and choose GitHub Credentials

**Configure**

- General
- Triggers
- Pipeline**
- Advanced

**Pipeline**

Define your Pipeline using Groovy directly or pull it from source control.

**Definition**

Pipeline script from SCM

SCM ?

Git

**Repositories ?**

Repository URL ?

https://github.com/Samiksha201104/MyMavenPracticeAutomationApp.git

Credentials ?

GitHub Credentials

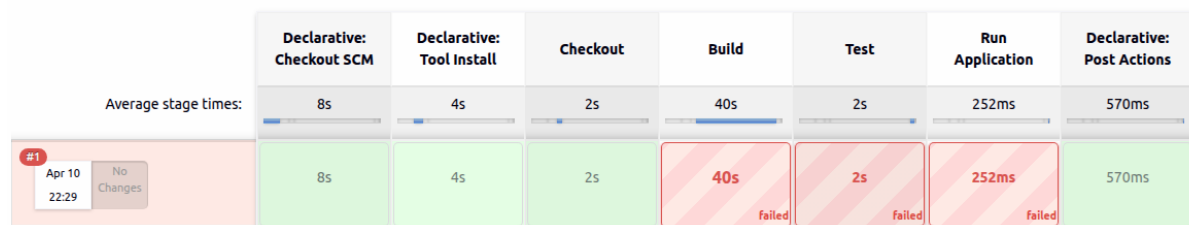
+ Add

**Step 7 :** in the Dashboard click on the MyMavenPracticeAutomationApp item and select Build Now and then Full Stage view

Dashboard view showing a list of pipelines. A context menu is open over the 'MyMavenPracticeAutomationApp' pipeline, showing options: Changes, Build Now, Configure, Delete Pipeline, Full Stage View, Stages, Rename, and Pipeline Syntax.

| S | W | Name                         | Duration | Last         | Duration |
|---|---|------------------------------|----------|--------------|----------|
| ✓ | ☁ | MyGradleApp                  | 1 min    | 5 min 51 sec | ▶        |
| ✓ | ☁ | MyMavenApp                   | 10 hr    | 33 sec       | ▶        |
| ⋮ | ☀ | MyMavenPracticeAutomationApp | N/A      | N/A          | ▶        |

#### MyMavenPracticeAutomationApp - Stage View



To correct the errors change the code in the pom.xml

`<source>1.7</source>`  
`<target>1.7</target>`

To

`<source>17</source>`  
`<target>17</target>`

Save and Build again in Jenkins



## MyMavenPracticeAutomationApp - Stage View

|                                                   |                             | Declarative:<br>Checkout SCM | Declarative:<br>Tool Install | Checkout | Build & Test | Run<br>Application | Declarative:<br>Post Actions |
|---------------------------------------------------|-----------------------------|------------------------------|------------------------------|----------|--------------|--------------------|------------------------------|
| Average stage times:<br>(full run time: ~2min 8s) |                             | 5s                           | 2s                           | 43s      | 51s          | 32s                | 600ms                        |
| #10                                               | Apr 12<br>19:07<br>1 commit | 3s                           | 175ms                        | 2s       | 34s          | 1min 11s           | 873ms                        |